

Shayan Bali

shayanbali@gmail.com | +44 7376 446018 | London, UK | [LinkedIn](#) | [GitHub](#)

SUMMARY

AI/ML researcher and engineer with a background in deep learning and NLP, and over 4 years experience in Python, TensorFlow, PyTorch, and other ML libraries. Interested in solving problems using AI, collaborating with teams, and staying up to date with recent advances in the field. Seeking a challenging role to develop cutting-edge AI solutions.

EDUCATION

M.Sc. in Advanced Computing

SEP 2024 – SEP 2025

King's College London, Total Grade: Distinction

London, United Kingdom

- Thesis: Making Visual Content Accessible: A Full-Stack Framework for Interactive Mind-Map Summarization Using AI and Large Language Models (LLMs). [\[GitHub\]](#)

B.Sc. in Computer Engineering

SEP 2019 – APR 2024

Amirkabir University of Technology, Total Grade: 18.60/20 (top 6%)

Tehran, Iran

- Thesis: Disease Diagnosis based on Brain Functional Network Analysis Focusing on Graph Neural Networks. [\[GitHub\]](#)

WORK EXPERIENCE

Machine Learning Engineer

MAY 2023 – SEP 2023

Vista Samaneh Asa, [\[LinkedIn\]](#)

Tehran, Iran

- Developed a multi-label text classification model for stock brokerage news categorization, achieving 93% accuracy using a BERT-based encoder.
- Built a Named Entity Recognition (NER) system to extract entities from financial news, reaching 95% token-level accuracy by using XLM-RoBERTa.
- Implemented a text summarization system for news articles, reducing text length by 30–60% by adapting the Google PageRank algorithm for sentence ranking.
- Collaborated with cross-functional teams to integrate AI models into real-time production systems, and deliver AI solutions for a stock brokerage with seven million users.

Artificial Intelligence Intern

JUL 2021 – OCT 2021

Faradis Alborz, [\[LinkedIn\]](#)

Tehran, Iran

- Implemented a predictive model for ATM cash demand using two years of data from 500+ ATMs with time-series models and neural networks in Keras, reducing forecast error by 12% via optimized window sizes and external economic indicators.

ACADEMIC EXPERIENCE

Research Assistant

AUG 2023 – SEP 2024

Institute for Research in Fundamental Sciences (IPM), [\[LinkedIn\]](#)

Tehran, Iran

- Investigated explainability in Vision Transformers and GANs by evaluating over 20K images with attention attribution (Grad-CAM, Attention Rollout) and latent space analysis, revealing key patterns in model interpretability.

Research Assistant

AUG 2023 – JUL 2024

Shahid Beheshti University, [\[LinkedIn\]](#)

Tehran, Iran

- Designed a transformer-based model to distinguish AI-generated from human-written text using semantic and syntactic features, attaining F1-scores of 0.853 (multilingual) and 0.858 (monolingual/English), demonstrating strong performance on AI text classification tasks.

Teaching Assistant

OCT 2023 – JAN 2024

Amirkabir University of Technology, [\[LinkedIn\]](#)

Tehran, Iran

- Mentored over 150 students across 6+ courses, including Data Mining, Algorithms, Applied Linear Algebra, and Data Structures, by crafting weekly assignments, grading projects, and recording tutorial videos to reinforce course concepts.

PROJECTS

Interactive Mind-map Summarization Framework

2025

Thesis Project, [\[GitHub\]](#)

Python, PyTorch, TypeScript, HTML, CSS, JavaScript

- Developed a full-stack framework with TypeScript and Python that converts audiovisual content into interactive mind-map summaries using ASR, topic modeling, segmentation, summarization, and TTS, with an accessible WCAG-compliant interface for users with complex communication needs.

UK Football Performance Visualizer

2025

Data Processing and Visualization, [\[GitHub\]](#)

Python, HTML, CSS, JavaScript

- Built a web-based interactive tool integrating 20 seasons (2005–2024) of football data to visualize regional performance of teams through choropleth maps and trend charts across UK counties.

Image Classification Using Transfer Learning

2024

Deep Learning Project, [\[GitHub\]](#)

PyTorch

- Evaluated the performance of pre-trained models (AlexNet, VGG16, ResNet) for image classification on the Natural Images dataset (6,899 images), achieving up to 98% test accuracy while analyzing model complexity-performance key trade-offs.

Game Recommendation System

2023

Cloud Computing and Big Data Project, [\[GitHub\]](#)

Python, Apache Spark

- Delivered personalized game recommendations across 10,000+ titles with 92% average coverage by utilizing collaborative filtering based on user ratings.

PUBLICATIONS

- Hosseini, M.*, Hosseini, K.*, **Bali, S.**, Zanjani, Z., Momtazi, S. "PerHalluEval: Persian Hallucination Evaluation Benchmark for Large Language Models". Submitted to LREC 2026. (*** equal contribution**) [\[arXiv\]](#)
- Farsi, F.*, Aghababaloo, F.*, Shariati Motlagh, S.*, Ghofrani, P., Sadraei Javaheri, M., **Bali, S.**, Shabani, A., Bijary, F., Zamaninejad, G., Salehoof, A., Momtazi, S. "MELAC: Massive Evaluation of Large Language Models with Alignment of Culture in Persian Language.". Accepted in IJCNLP-AAACL (2025). [\[arXiv\]](#)
- Farsi, F., Shariati Motlagh, S., **Bali, S.**, Sabouri, S., Momtazi, S. "Persian in a Court: Benchmarking VLMs in Persian Multi-Modal Tasks". Proc. 1st Workshop on Evaluation of Multimodal Generation @ COLING 2025. [\[ACL Anthology\]](#)
- Rad, M.*, Farsi, F.*, **Bali, S.***, Etezadi, R., Shamsfard, M. "Investigating Syntactic and Semantic Features for Distinguishing AI-Generated and Human-Written Texts". Proc. 18th SemEval @ NAACL 2024. [\[ACL Anthology\]](#)

SKILLS

- **Language:** English (Fluent, IELTS Overall Score: 7.5; R : 8; L : 8.5; S : 7.5; W : 6.5), Persian (Native)
- **Programming Languages:** Python, Java, C, JavaScript, TypeScript
- **ML/AI Technologies:** PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, LLM APIs, Natural Language Processing (NLP), Signal Processing, Machine Learning, Deep Learning
- **Dev Tools & Platforms:** Git, Docker, Kubernetes, FastAPI, Apache Spark
- **Data & Analysis:** SQL, Pandas, NumPy, Matplotlib

EXTRACURRICULAR ACTIVITIES

Workshop Instructor at Amirkabir Artificial Intelligence Student Summit (AAISS)

JAN 2025

- Delivered an introductory workshop on Natural Language Processing (NLP), covering core concepts, deep learning methods, common tasks, and popular libraries for beginner students.

Presenter at 31st International Conference on Computational Linguistics (COLING), Abu Dhabi, UAE

JAN 2025

- Presented the accepted paper : "Persian in a Court: Benchmarking VLMs in Persian Multi-Modal Tasks".

Member of Computer Engineering Student Council, Amirkabir University of Technology

JUL 2021 – Nov 2022

- Collaborated with council members to organize student events, represent concerns, and improve curriculum and facilities.